Spectral Simulation Report

Automated Analysis

June 18, 2025

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1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot and a detailed table summarizing the initial composition and actual labeled positions of elements, along with simulation parameters such as temperature and electron density.

2 Sample: 16197

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	9040 K
Electron Density (n_e)	7.92e+15 $\mathrm{cm}^{-}3$

Table 1: Simulation parameters for sample 16197.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Labeled Position (%)**
Al	10.87	10.74
Ar	0.00	0.00
C	0.00	0.00
Ca	6.83	3.66
Cl	7.15	0.66
Co	6.13	34.28
Cr	10.34	9.69
Fe	0.00	0.00
Mg	10.49	8.47
Mn	11.31	17.24
N	5.84	9.69
Na	3.75	7.86
Ni	3.52	11.67
O	7.31	4.05
S	7.85	5.05
Si	8.63	5.44
Ti	0.00	0.00
background	N/A	28.64
Total (%)		157.15

Table 2: Composition analysis for sample 16197.

2.3 Spectral Plot

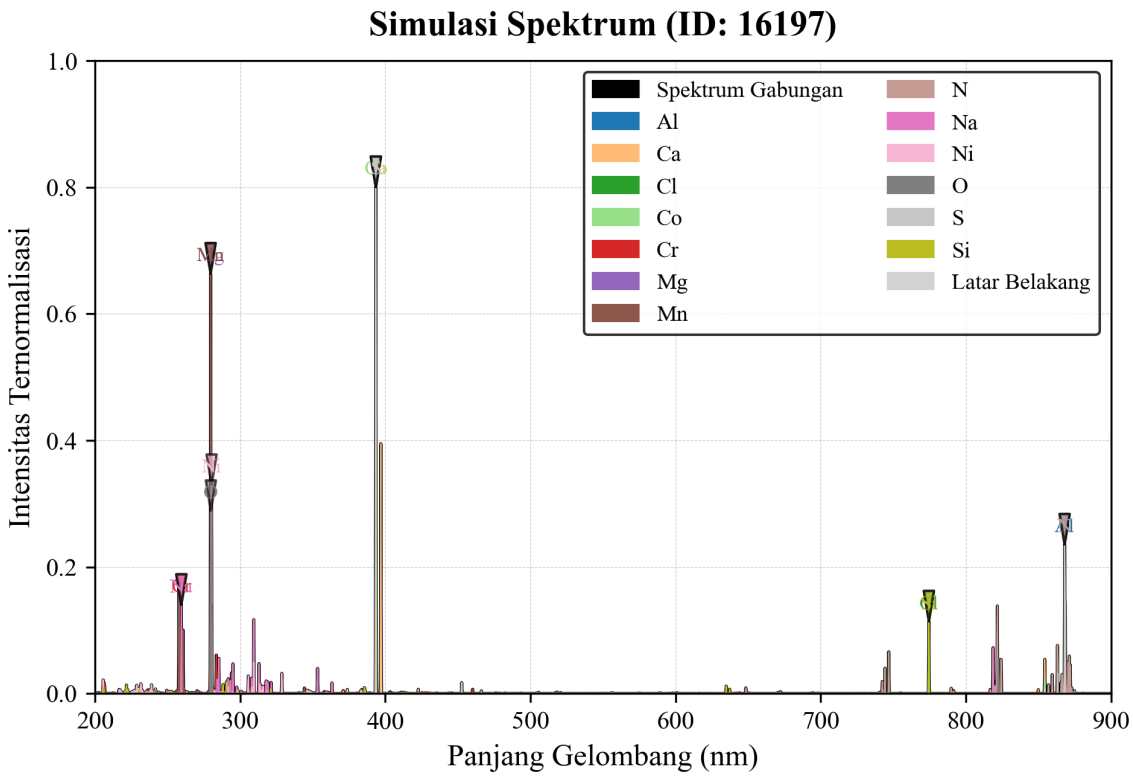

Figure 1: Spectral simulation plot for sample 16197.

3 Sample: 5639

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	10556 K
Electron Density (n_e)	3.13e+15 $\mathrm{cm}^{-}3$

Table 3: Simulation parameters for sample 5639.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	0.00	0.00
Ar	0.00	0.00
C	0.00	0.00
Ca	5.56	3.66
Cl	14.21	0.85
Co	11.98	34.28
Cr	8.58	9.69
Fe	0.00	0.00
Mg	4.15	8.57
Mn	3.40	17.24
N	2.14	9.89
Na	2.89	7.86
Ni	12.13	11.67
O	10.06	4.79
S	10.09	5.57
Si	14.81	5.44
Ti	0.00	0.00
background	N/A	31.45
Total (%)		150.95

Table 4: Composition analysis for sample 5639.

3.3 Spectral Plot

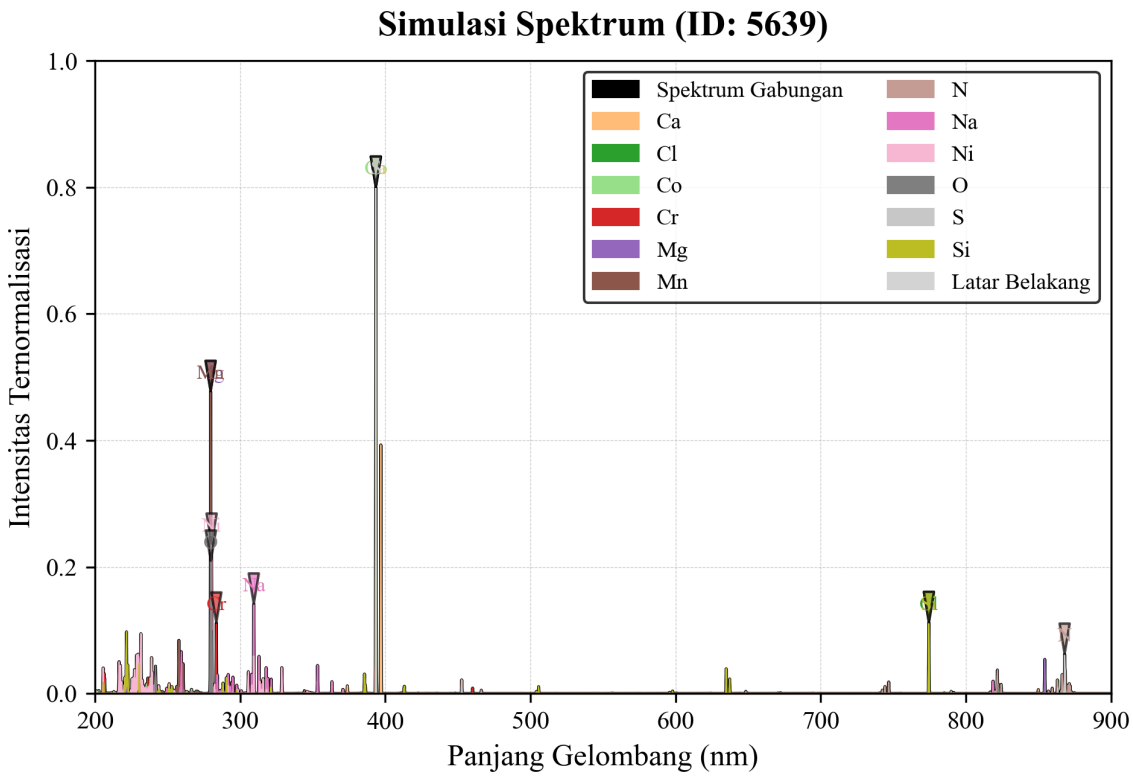

Figure 2: Spectral simulation plot for sample 5639.

4 Sample: 2441

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	9242 K
Electron Density (n_e)	1.87e+17 $\mathrm{cm}^{-}3$

Table 5: Simulation parameters for sample 2441.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	0.65	9.35
Ar	5.52	17.70
C	6.59	5.66
Ca	8.22	3.66
Cl	8.81	0.39
Co	10.22	34.28
Cr	2.34	9.69
Fe	0.00	0.00
Mg	9.61	8.54
Mn	10.82	17.24
N	0.84	9.67
Na	1.53	8.06
Ni	10.73	11.67
O	7.75	4.15
S	5.03	3.71
Si	4.22	5.32
Ti	7.10	28.96
background	N/A	19.02
Total (%)		197.07

Table 6: Composition analysis for sample 2441.

4.3 Spectral Plot

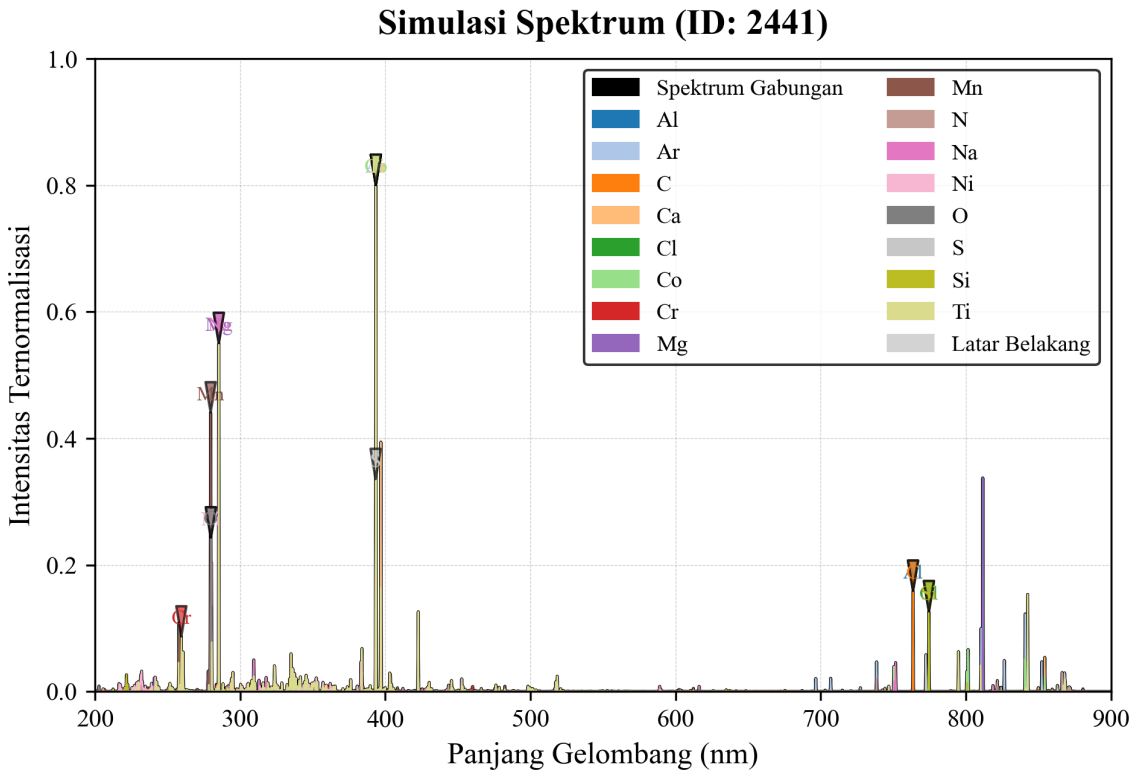

Figure 3: Spectral simulation plot for sample 2441.

5 Sample: 1644

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	7424 K
Electron Density (n_e)	1.12e+15 $\mathrm{cm}^{-}3$

Table 7: Simulation parameters for sample 1644.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	2.66	5.32
Ar	6.73	17.48
C	2.49	5.40
Ca	13.53	3.66
Cl	0.84	0.29
Co	0.00	0.00
Cr	11.87	9.69
Fe	0.00	0.00
Mg	10.88	7.74
Mn	12.83	17.24
N	0.00	0.00
Na	10.87	7.86
Ni	6.46	11.67
O	8.89	2.20
S	0.84	0.61
Si	5.56	5.32
Ti	5.56	28.88
background	N/A	33.28
Total (%)		156.64

Table 8: Composition analysis for sample 1644.

5.3 Spectral Plot

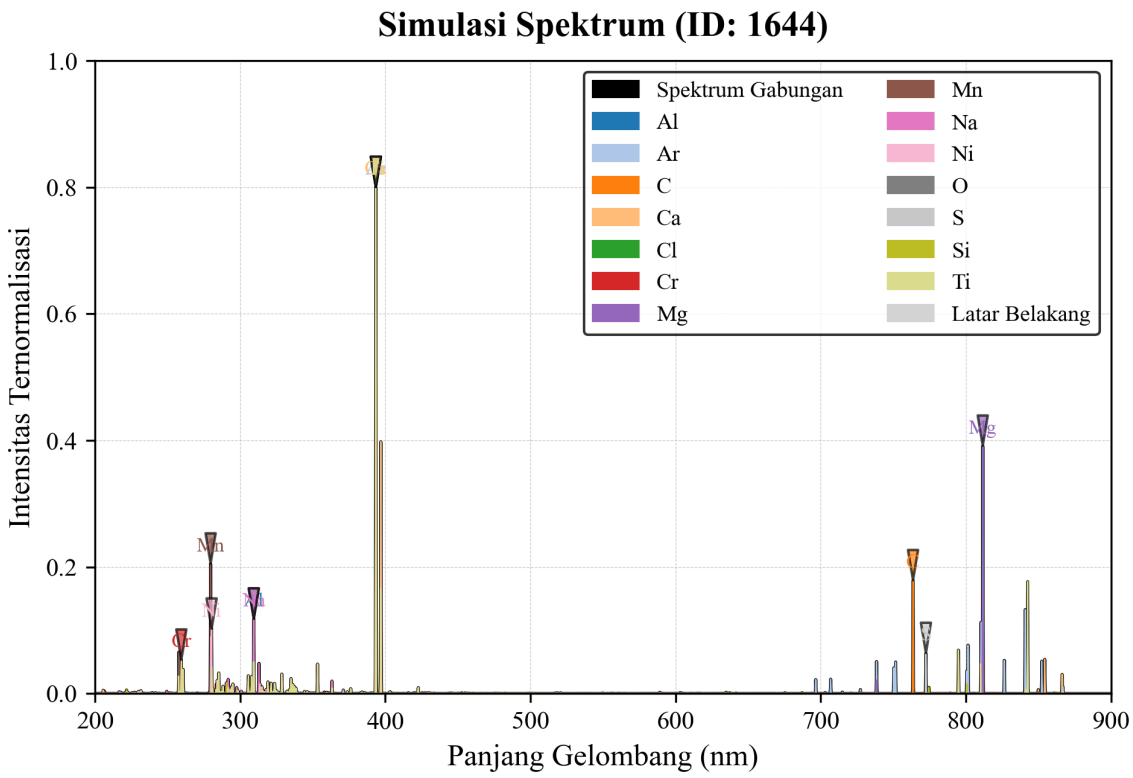

Figure 4: Spectral simulation plot for sample 1644.

6 Sample: 11151

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	5707 K
Electron Density (n_e)	1.23e+15 $\mathrm{cm}^{-}3$

Table 9: Simulation parameters for sample 11151.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	6.92	3.32
Ar	9.33	6.81
C	6.83	2.88
Ca	7.40	3.66
Cl	10.20	0.29
Co	6.52	20.70
Cr	9.55	9.69
Fe	3.55	52.08
Mg	3.80	3.76
Mn	8.12	15.50
N	0.00	0.00
Na	1.21	7.86
Ni	4.07	11.67
O	7.44	0.20
S	10.94	0.61
Si	2.25	3.81
Ti	1.86	28.83
background	N/A	26.66
Total (%)		198.34

Table 10: Composition analysis for sample 11151.

6.3 Spectral Plot

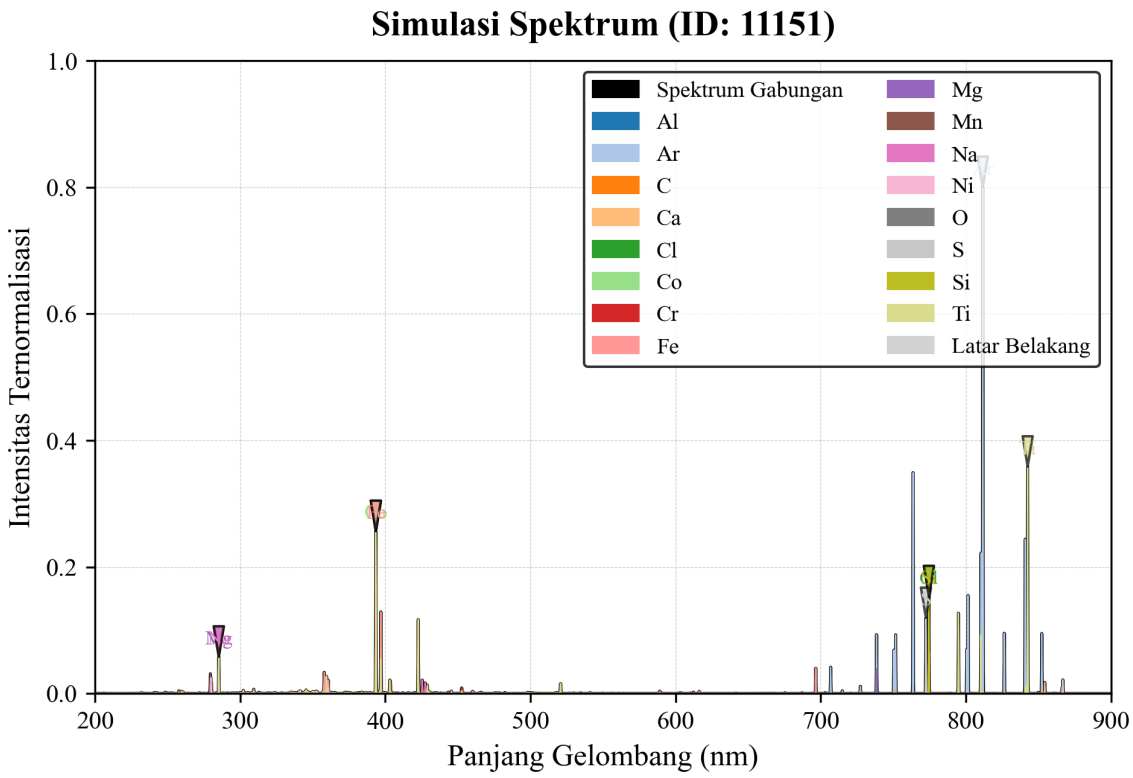

Figure 5: Spectral simulation plot for sample 11151.